mztabm2cytoscape: Pathway Mapping and Visualization of Metabolomics Data

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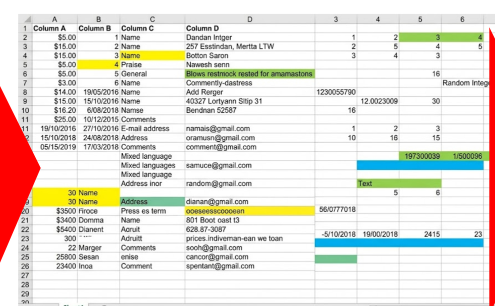
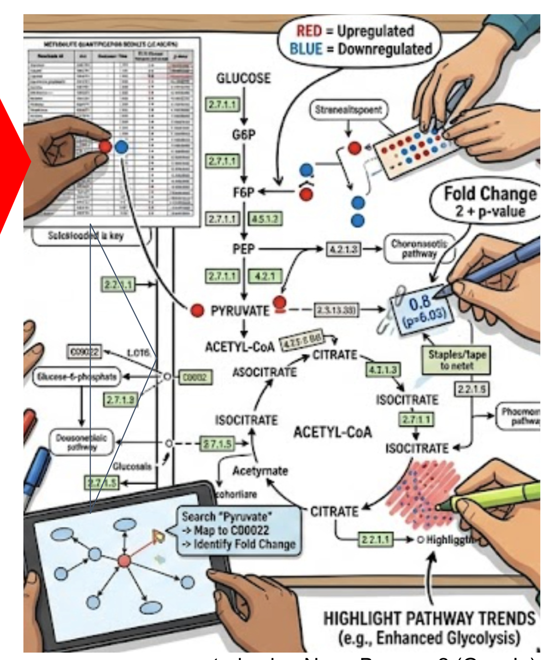
mztabm2cytoscape is a workflow implementation automating pathway mapping and visualization of MS quantitative values, a representative downstream analysis in MS based metabolomics.

Background & Problem

generated using Nano Banana 2 (Google)



generated using Nano Banana 2 (Google)



generated using Nano Banana 2 (Google)

Quantitative data output from a wide variety of upstream MS analysis software

Unstructured spreadsheet data

- Different data formats across upstream MS analysis software
- Downstream analysis software ends up having to implement separate workflows for every different data format.

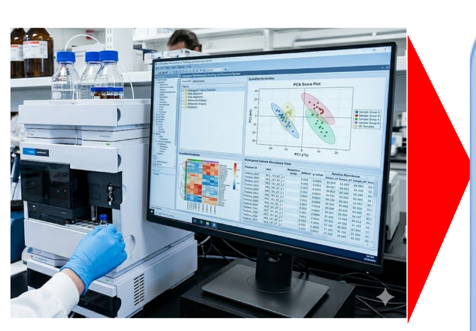
Manual mapping

- Mapping small molecule identifiers between "MS quantitative data" and "biological pathways"
- Mapping small molecule quantitative value visualization (e.g., box plots) onto biological pathways

The challenge

Implementing a streamlined, automated workflow for pathway mapping and visualization, ensuring seamless and consistent use of MS quantitative data from any upstream analysis software.

What we developed



Workflow logic

1. Standardization

Standardizing workflow inputs on mzTab-M achieves a consistent downstream analysis implementation.

2. Plotting

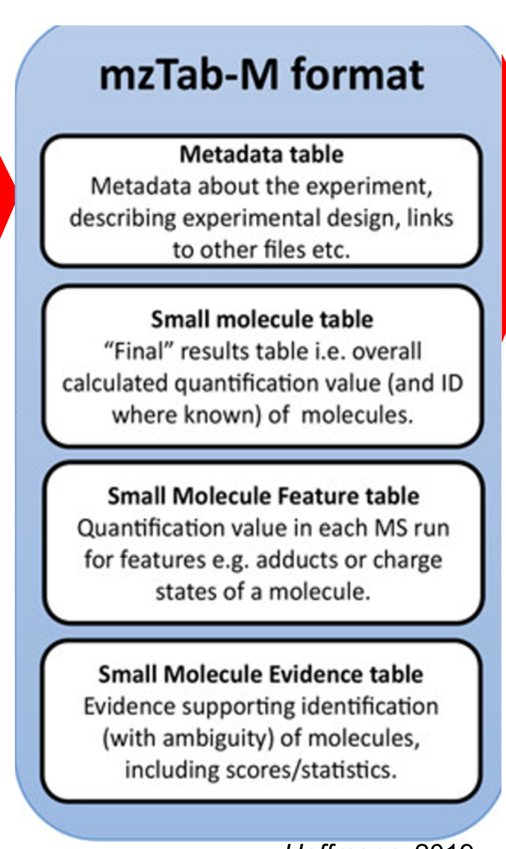
Using general-purpose plotting libraries (e.g., matplotlib) allows for flexible and complex visualization of MS quantitative data.

3. Harmonization

Each small molecule quantitative value is linked to a pathway node.

4. Network integration

Cytoscape displays the quantitative values on a pathway network.



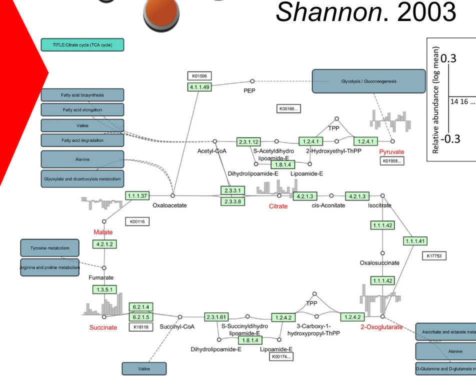
Hoffmann. 2019

mzTab-M

A HUPO-PSI data standard for sharing quantitative results in MS based metabolomics

Cytoscape

Shannon. 2003



Fukushima. 2014

Cytoscape

network visualization software with flexible data integration. Also equipped with automation support features. (Otasek. 2019)

The goal

Versatile and reproducible pathway projection

for interpreting quantitative MS metabolomics

Methods

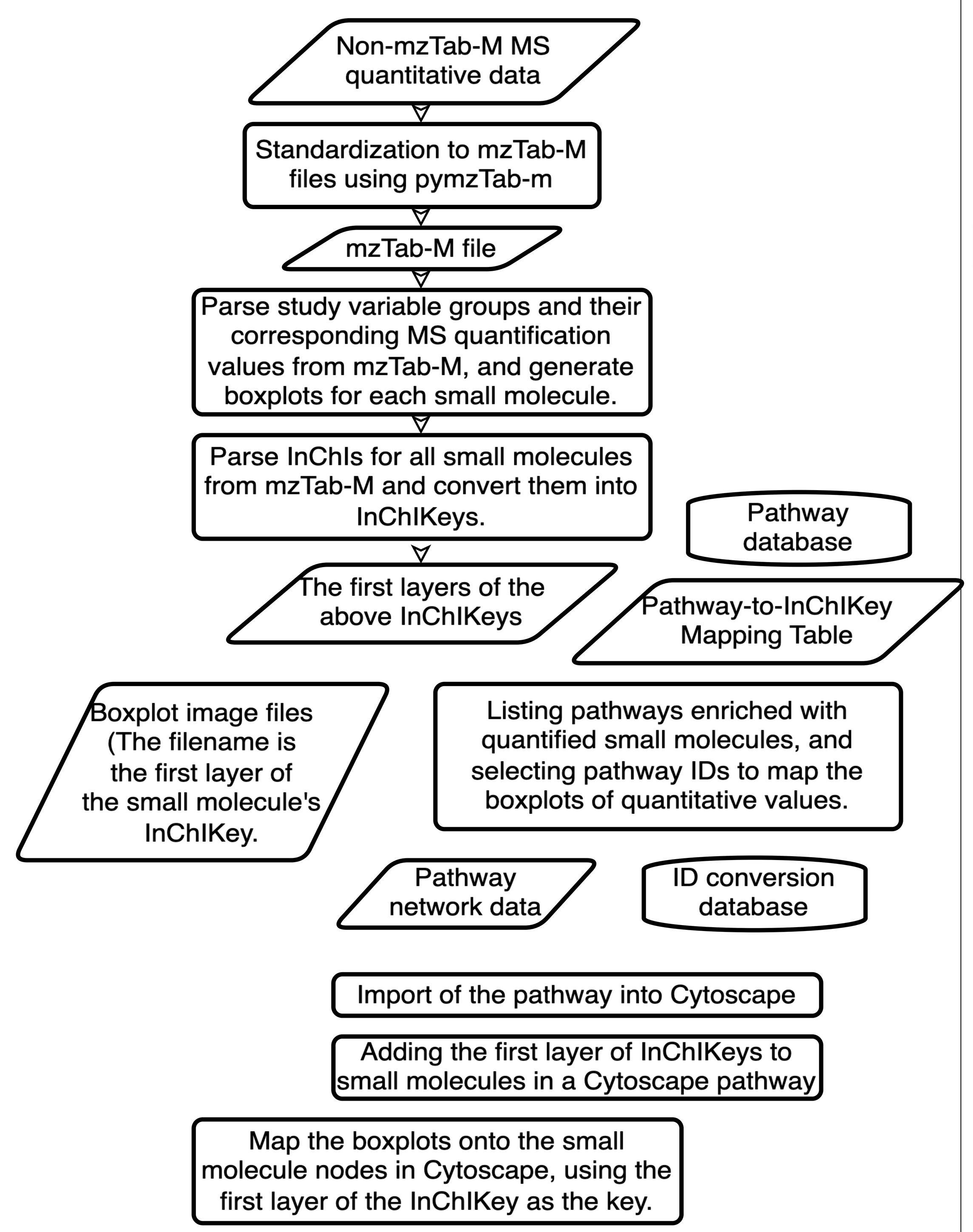


Figure 1.

Results

The pathway visualization on the right was generated from the data below via the mztabm2cytoscape workflow.

- MTBLS40 ISA-TABs from MetaboLights
- bbb

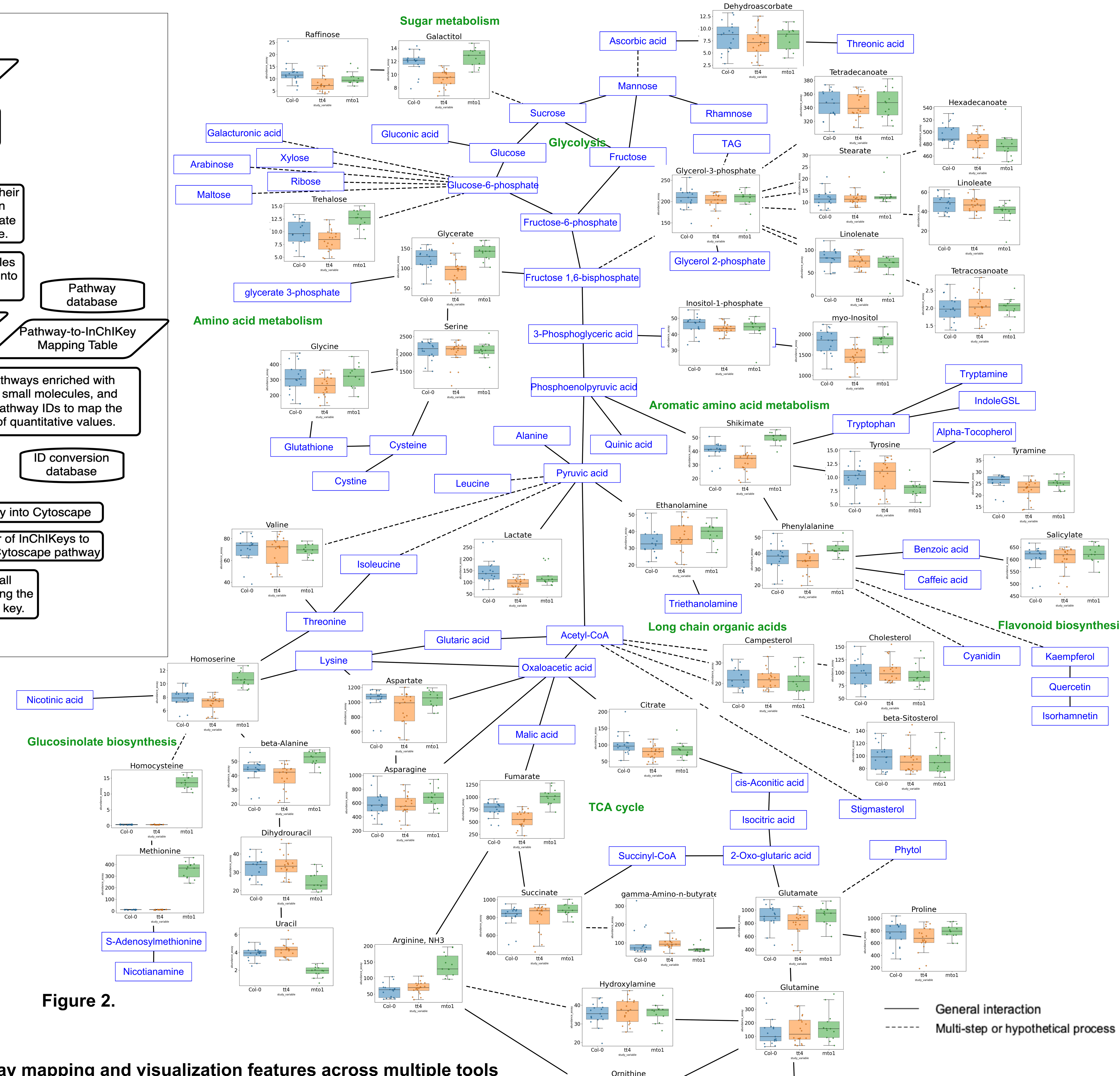


Figure 2.

Discussions

Table 1. Comparison of pathway mapping and visualization features across multiple tools

	mztabm2cytoscape	MetaboAnalyst	VANTED	pathview
Use of mzTab-M for data integration with pathways	✓			
Data integration with any pathway data source	✓	KEGG only	✓	KEGG only
Automation of the workflow	✓	✓		
How small molecule IDs are standardized / mapped	First block of InChIKey	Its own internal ID	Arbitrary ID (Manual standardization required)	KEGG ID (Manual standardization required)
Visuals that can be mapped on the pathway nodes	Complex plots, such as boxplots, or anything that can be generated with matplotlib.	Color scale only	Simple bar graph or line chart	Color scale only

This project's repository, its dependencies, and the data used.

- https://github.com/kozo2/mztabm2cytoscape
- https://github.com/lifs-tools/pymzTab-m
- https://matplotlib.org
- https://pandas.pydata.org/
- https://www.rdkit.org
- https://cytoscape.org/
- https://identifiers.org/metabolights:MTBLS40
- https://identifiers.org/wikipathways:WP3622